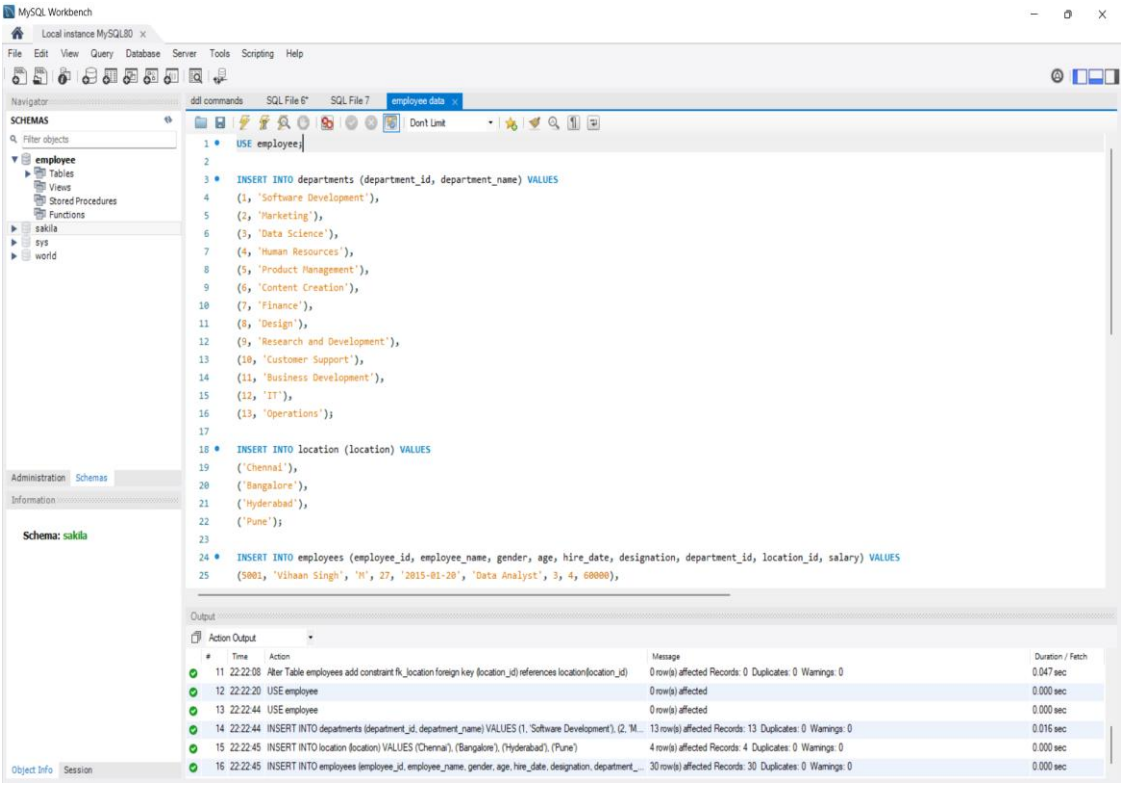
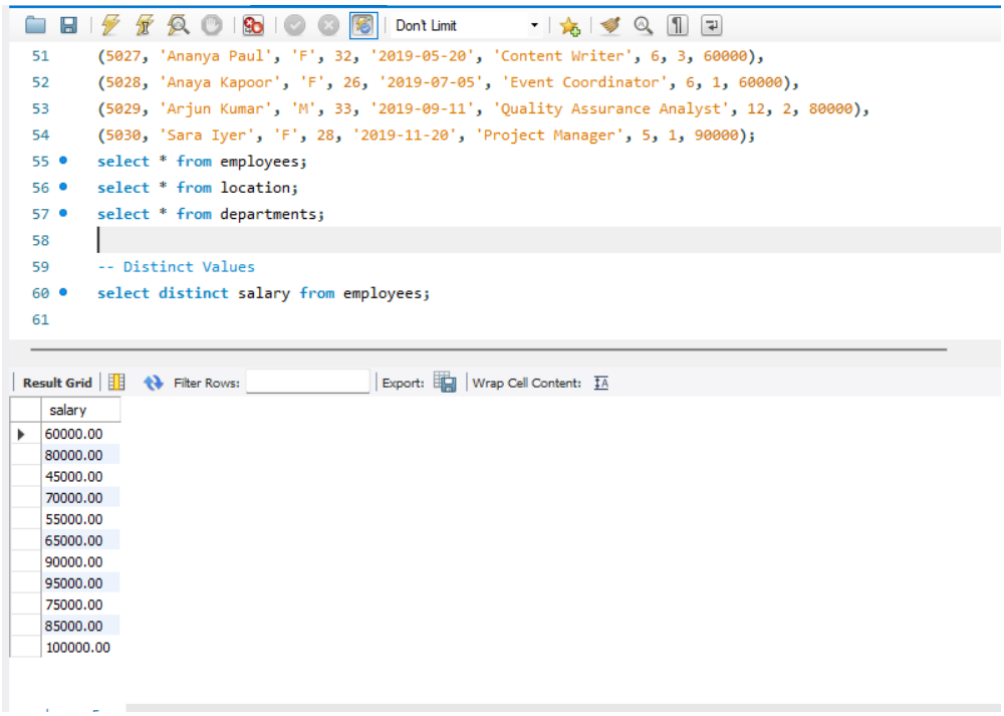
MySQL Assignment 2 – Querying Data

DML insert codes



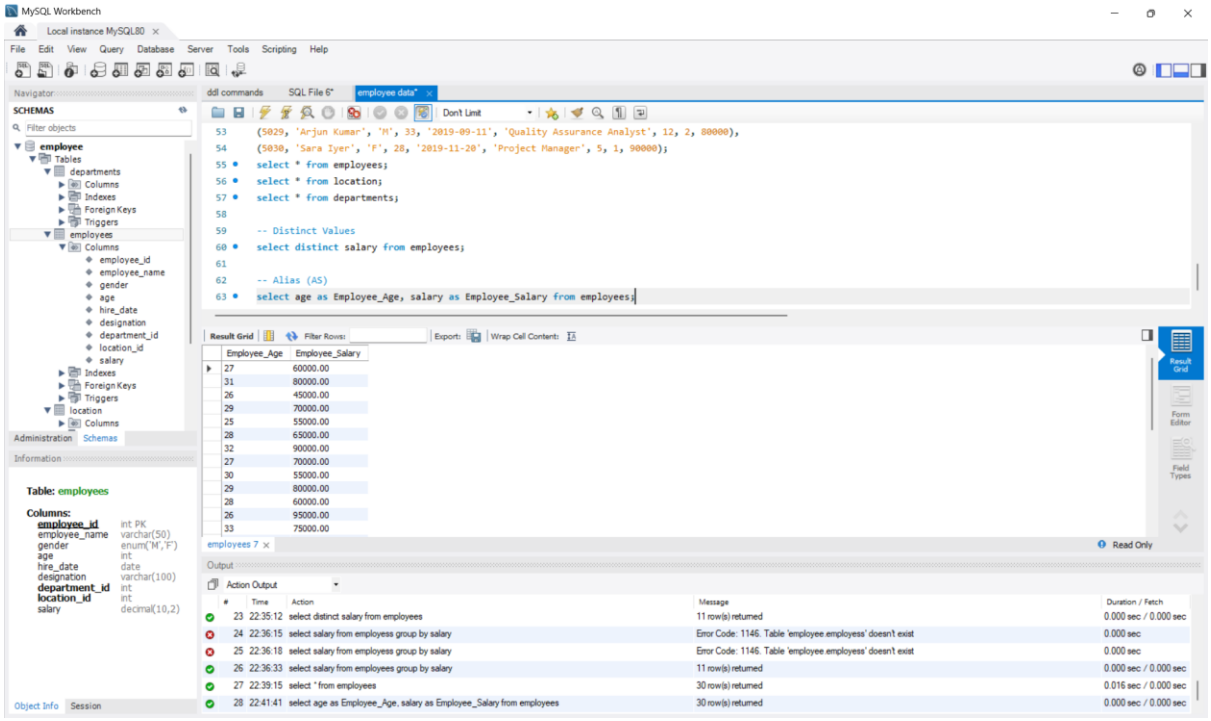
Distinct Values:

◉ a query to retrieve distinct salaries from the Employees table.



Alias (AS):

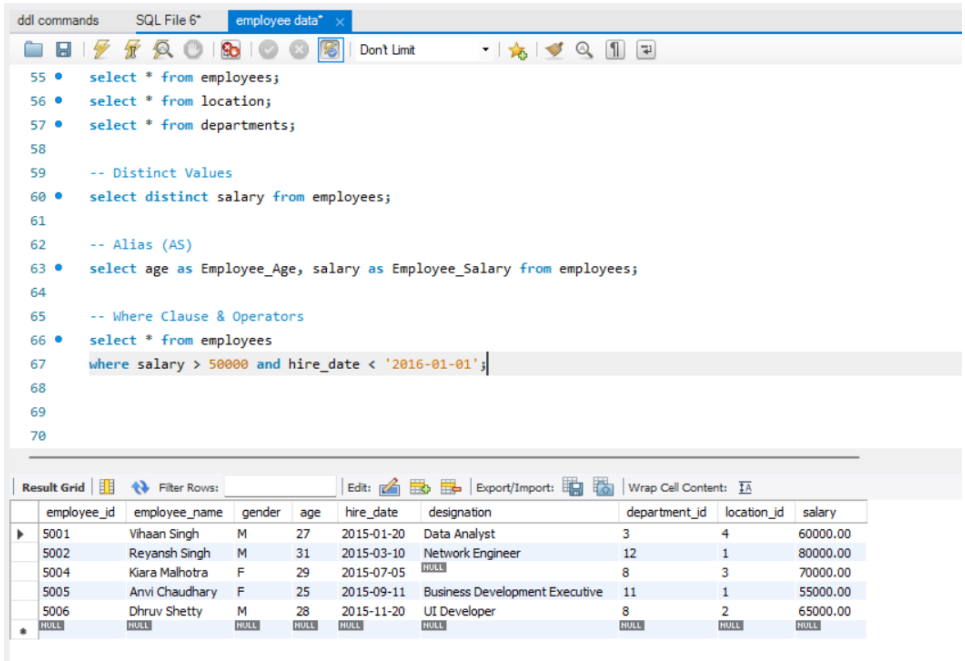
◉ Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively



Where Clause & Operators:

◉ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

◉ Find the employee whose designation is missing and fill it with "Data Scientist".



SQL File 6* | mysql assignment 2*

Don't Limit

```
10
11      -- Where Clause & Operators
12  •   select * from employees where salary > 50000 and hire_date < '2016-01-01';
13  •   update Employees
14      set designation = 'Data Scientist'
15      where designation IS NULL;
16  •   select * from employees where designation ='Data Scientist';
17
18      -- Sorting and Grouping Data
19      -- ORDER BY
20  •   select * from employees order by department_id asc, salary desc;
```

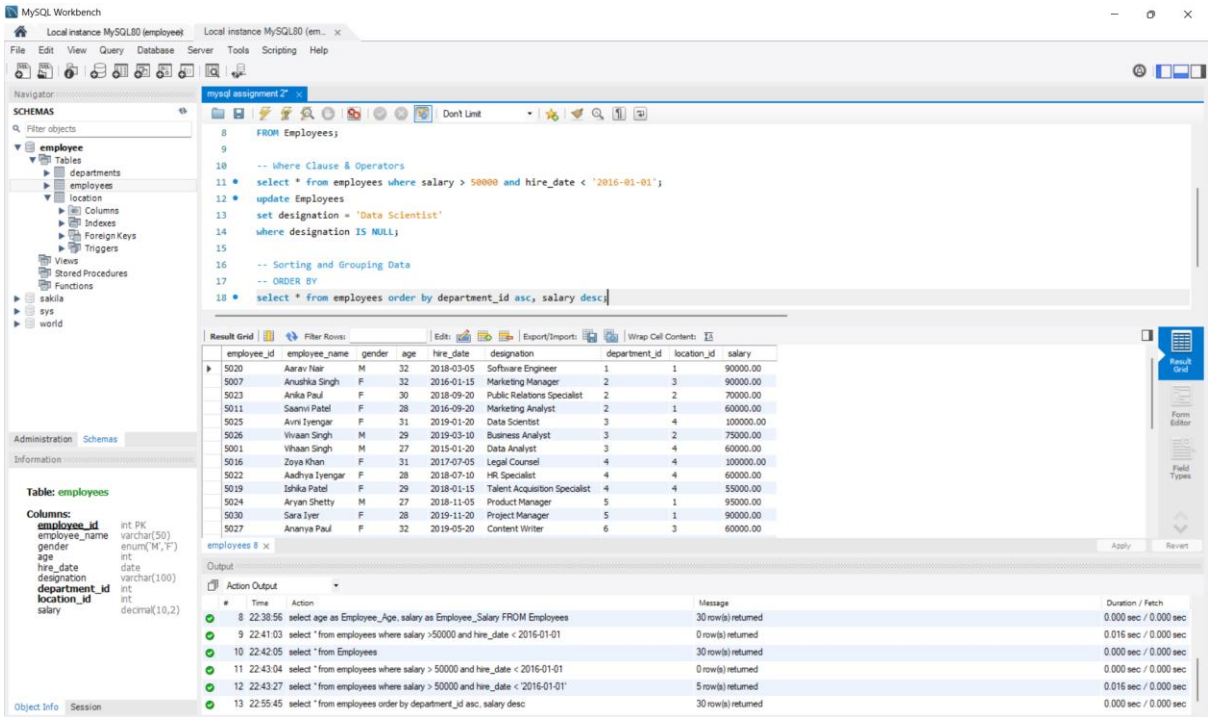
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

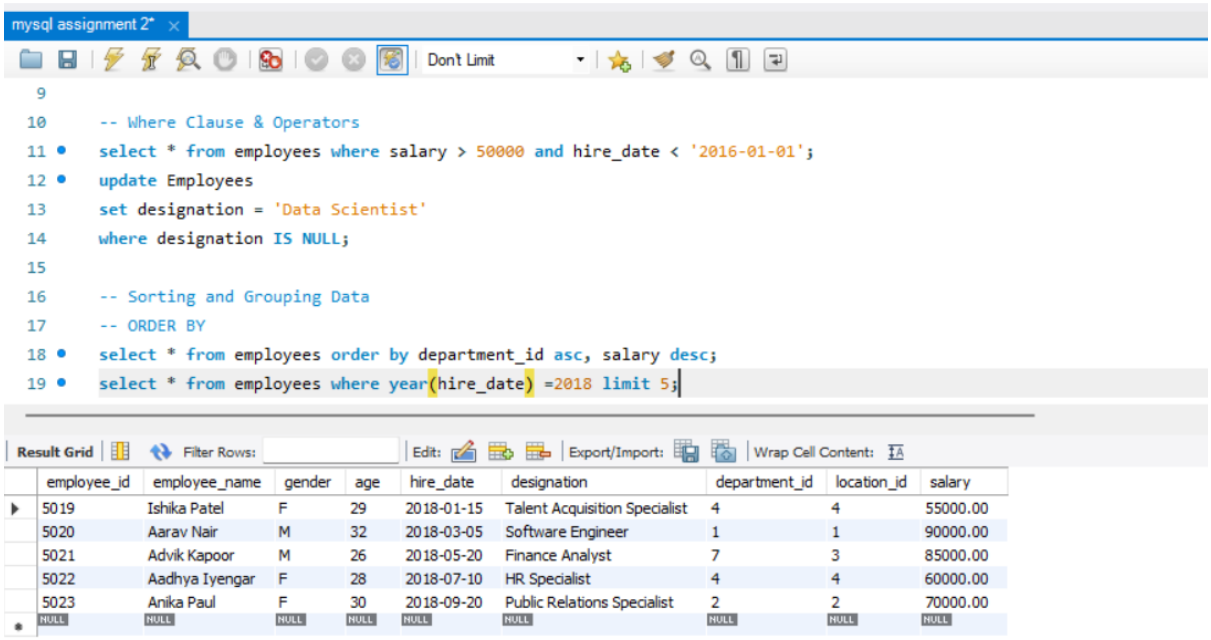
employees 3

Sorting and Grouping Data

◉ Find employees sorted by department ID in ascending order and salary in descending order.



Display the first 5 employees hired in the year 2018.

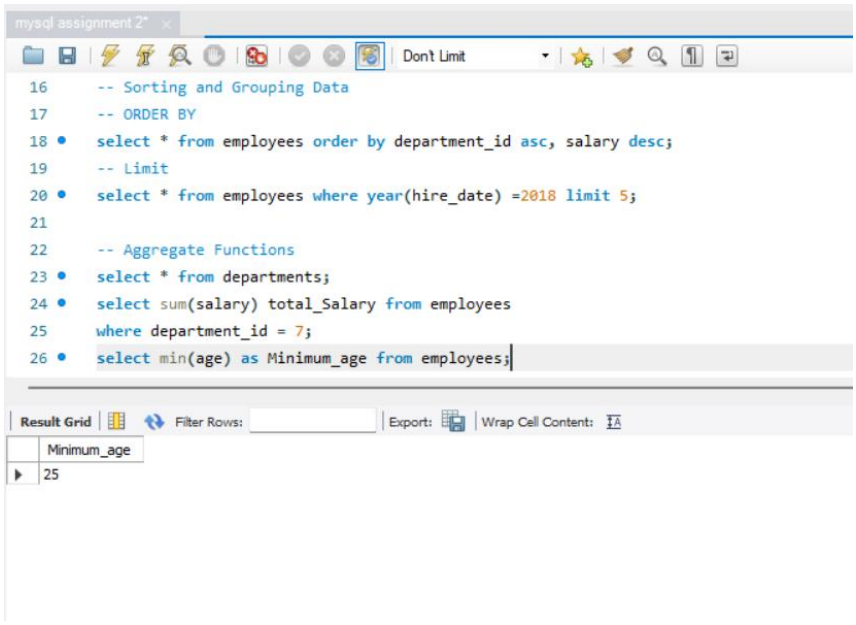




Aggregate Functions:

◉ Calculate the sum of all salaries in the Finance department.

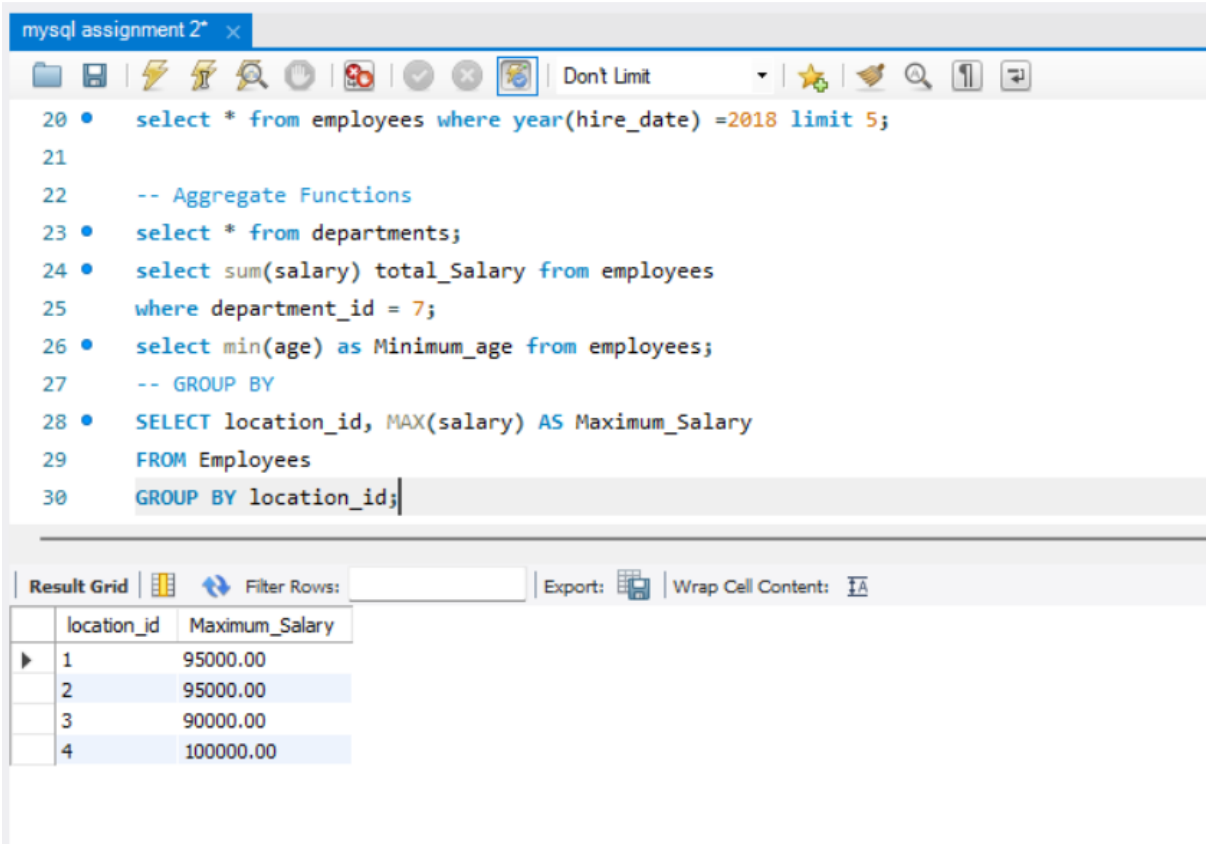
◉ Find the minimum age among all employees.

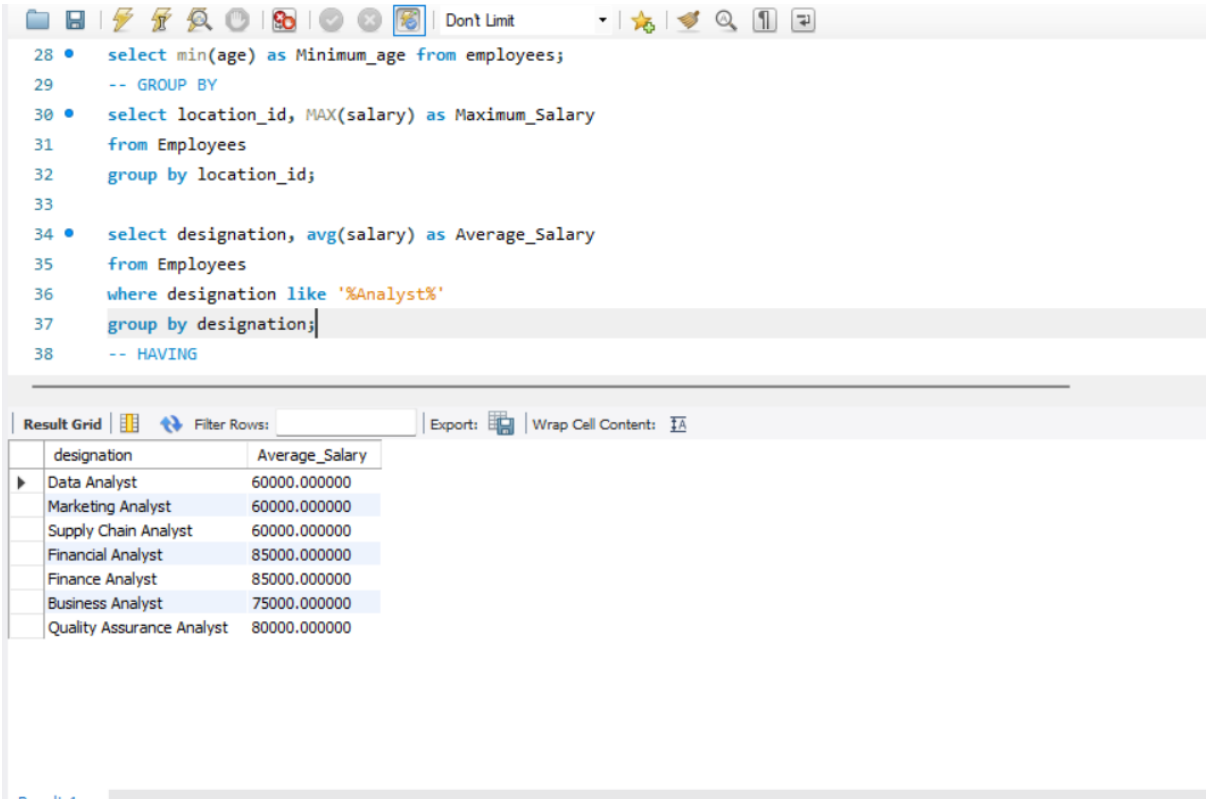


GROUP BY:

◉ List the maximum salary for each location.

◉ Calculate the average salary for each designation containing the word 'Analyst'.





HAVING:

◉ Find departments with less than 3 employees.

◉ Find locations with female employees whose average age is below 30.

```
23 •  select * from departments;
24 •  select sum(salary) total_Salary from employees
25    where department_id = 7;
26 •  select min(age) as Minimum_age from employees;
27    -- GROUP BY
28 •  SELECT location_id, MAX(salary) AS Maximum_Salary
29    FROM Employees
30    GROUP BY location_id;
31    -- HAVING
32 •  select department_id, count(employee_id) as employee_count from employees
33    group by department_id having employee_count < 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	employee_count
1	1
5	2
6	2
7	2
9	1
11	2
13	1

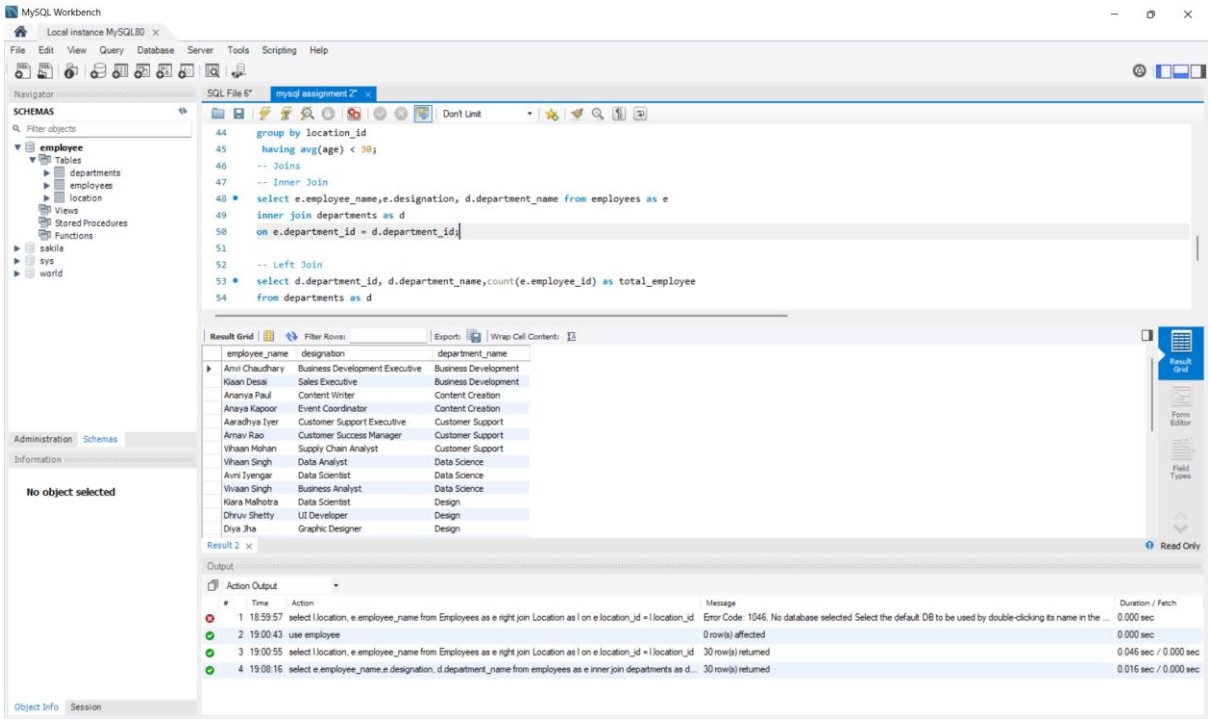
```
34    where designation like '%Analyst%'
35    group by designation;
36    -- HAVING
37 •  select department_id, count(employee_id) as employee_count from employees
38    group by department_id having employee_count < 3;
39
40 •  select location_id, avg(age) as Average_Age
41    from Employees
42    where gender = 'F'
43    group by location_id
44    having avg(age) < 30;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	Average_Age
1	26.7500
2	27.3333
4	29.2000

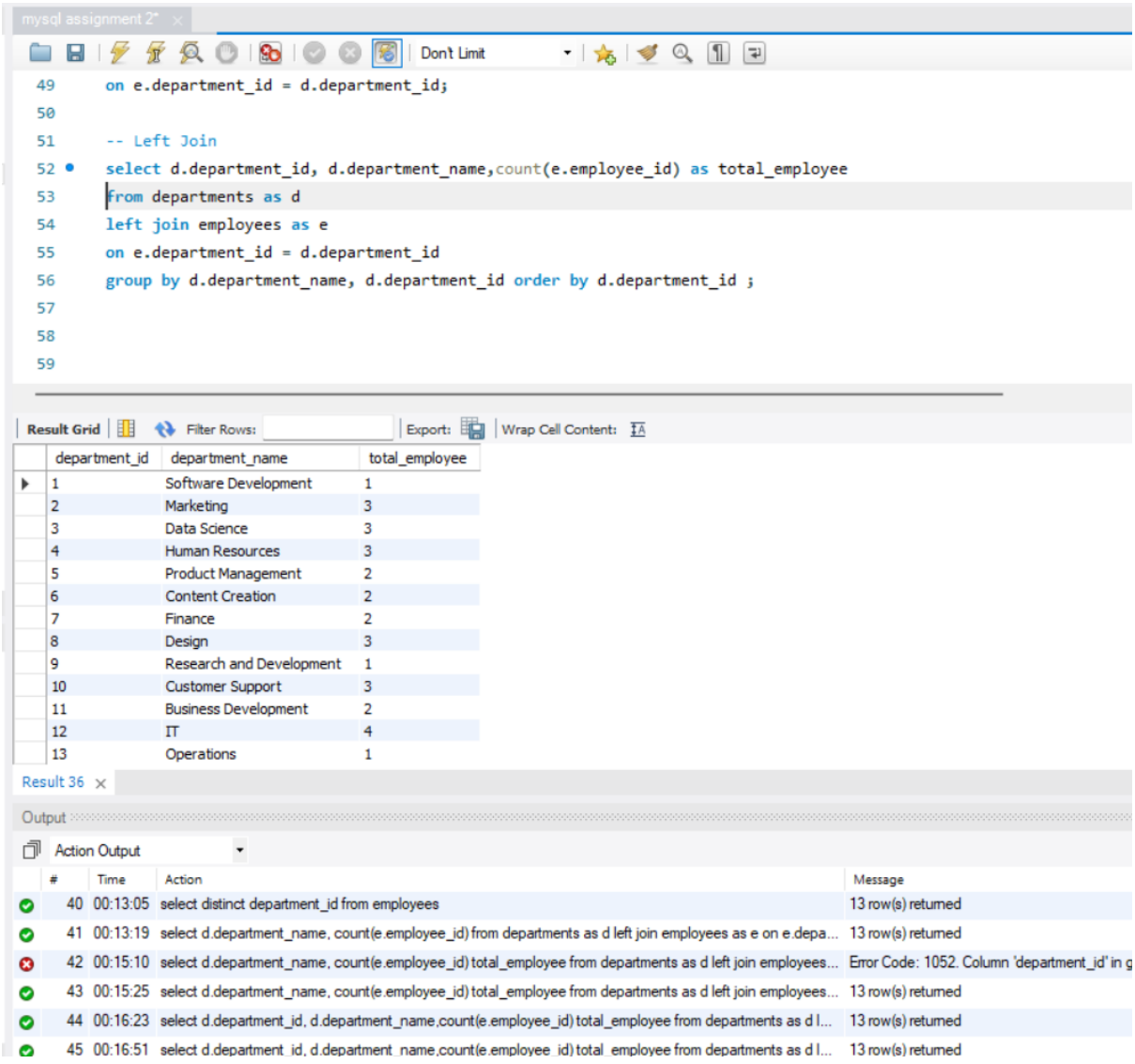
Inner Join:

List employee names, their designations, and department names where employees are assigned to a department



Left join

List all departments along with the total number of employees in each department, including departments with no employees.



```
mysql assignment 2*
Don't Limit
49    on e.department_id = d.department_id;
50
51    -- Left Join
52    select d.department_id, d.department_name,count(e.employee_id) as total_employee
53    from departments as d
54    left join employees as e
55    on e.department_id = d.department_id
56    group by d.department_name, d.department_id order by d.department_id ;
57
58
59
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	department_name	total_employee
1	Software Development	1
2	Marketing	3
3	Data Science	3
4	Human Resources	3
5	Product Management	2
6	Content Creation	2
7	Finance	2
8	Design	3
9	Research and Development	1
10	Customer Support	3
11	Business Development	2
12	IT	4
13	Operations	1

Result 36

Output

Action Output

#	Time	Action	Message
40	00:13:05	select distinct department_id from employees	13 row(s) returned
41	00:13:19	select d.department_name, count(e.employee_id) from departments as d left join employees as e on e.depa...	13 row(s) returned
42	00:15:10	select d.department_name, count(e.employee_id) total_employee from departments as d left join employees...	Error Code: 1052. Column 'department_id' in g
43	00:15:25	select d.department_name, count(e.employee_id) total_employee from departments as d left join employees...	13 row(s) returned
44	00:16:23	select d.department_id, d.department_name,count(e.employee_id) total_employee from departments as d l...	13 row(s) returned
45	00:16:51	select d.department_id, d.department_name,count(e.employee_id) total_employee from departments as d l...	13 row(s) returned

Right Join

Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

